Bootstrapping, Bagging, and Ensembling

GENERAL ASSEMBLY

Agenda

- Inference and the Central Limit Theorem (CLT)
- The bootstrapping method
- **B**ootstrap **agg**regat**ing** (bagging) decision trees
- General ensemble methods for models
- Python!

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What is inference?

The Central Limit Theorem (in a nutshell)

If you were to:

- *collect many repeated samples from some population,*
- *and you recorded the sample mean for each sample,*

then, in the long run, the distribution of the sample means would be Normal.

The Central Limit Theorem (in math)

This is often written:

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1)$$

For large *n*, we can write the above (approximately) as:

$$\bar{X} \approx N\left(\mu, \sigma/\sqrt{n}\right)$$

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Pause and Ponder

The CLT is only concerned with means.

What would a CLT for other statistics look like?

Does the population *median* have a defined long-run distribution for an arbitrary iid set of *X*s?

Pause and Ponder

The CLT is only concerned with means.

What would a CLT for other statistics look like?

Does the population *median* have a defined long-run distribution for an arbitrary iid set of *X*s?

Actually, no!

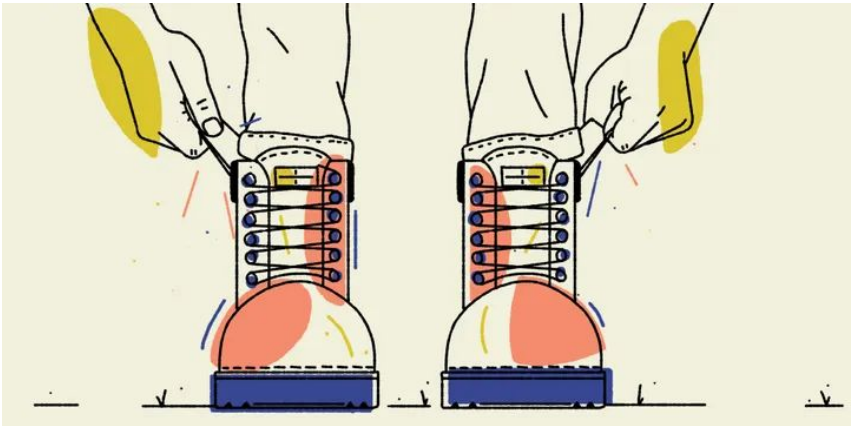
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Bootstrapping

The CLT fails for other statistics

What distribution does the population median have? The population standard deviation? The population 63rd percentile? The population regression coefficient?

We can *approximate* these things with **bootstrapping**.



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The Bootstrapping Technique

Bootstrapping requires 3 things:

- A sample size *n*
- A statistic of interest *t* (e.g. the median)
- A large number *B* (often a few hundred)

The Bootstrapping Technique

Given a sample size *n*, statistic *t*, and large number *B*:

1. Create a random sample from your data of size *n* **with replacement**
2. Compute the statistic *t* from this data and write it down
3. Perform steps 1-2 *B* times
4. You now have a set of sample statistics t_1, ..., t_B. This is your **bootstrapped distribution** and represents your estimate of the distribution of the statistic *t*

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Let's try it out!



The Wisdom of the Crowds

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Individuals are dumb, but crowds are smart!

It's an actual statistical fact that, instead of relying on one "ok" model's opinion, you'll get better results if you aggregate many "ok" models together!

Poll: How much does Penelope weigh, in pounds?



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This act of **ensembling** via **bagging** reduces the overall variance of your model. Since decision trees tend to overfit, they're a prime opportunity for bagging.

While less effective, bagging "dumb" learners (ie, underfit models) also tends to improve model fit! Win-win!

Bagging

We can simulate the wisdom of the crowds to improve our “simple” decision trees. The method of **b**ootstrap **agg**regat**ing**, or **bagging**, works as follows:

1. Create a bootstrap sample of our data
2. Fit a model M (in our case a decision tree)
3. Repeat steps 1-2 B times
4. You now have a set of models M_1, ..., M_B. Predictions are made as follows:

Bagging Prediction

For a given sample, compute its predictions for each model M_i.

For **regression**, the final prediction is the **mean** of these predictions.

For **classification**, the final prediction is the **majority vote** of these predictions.

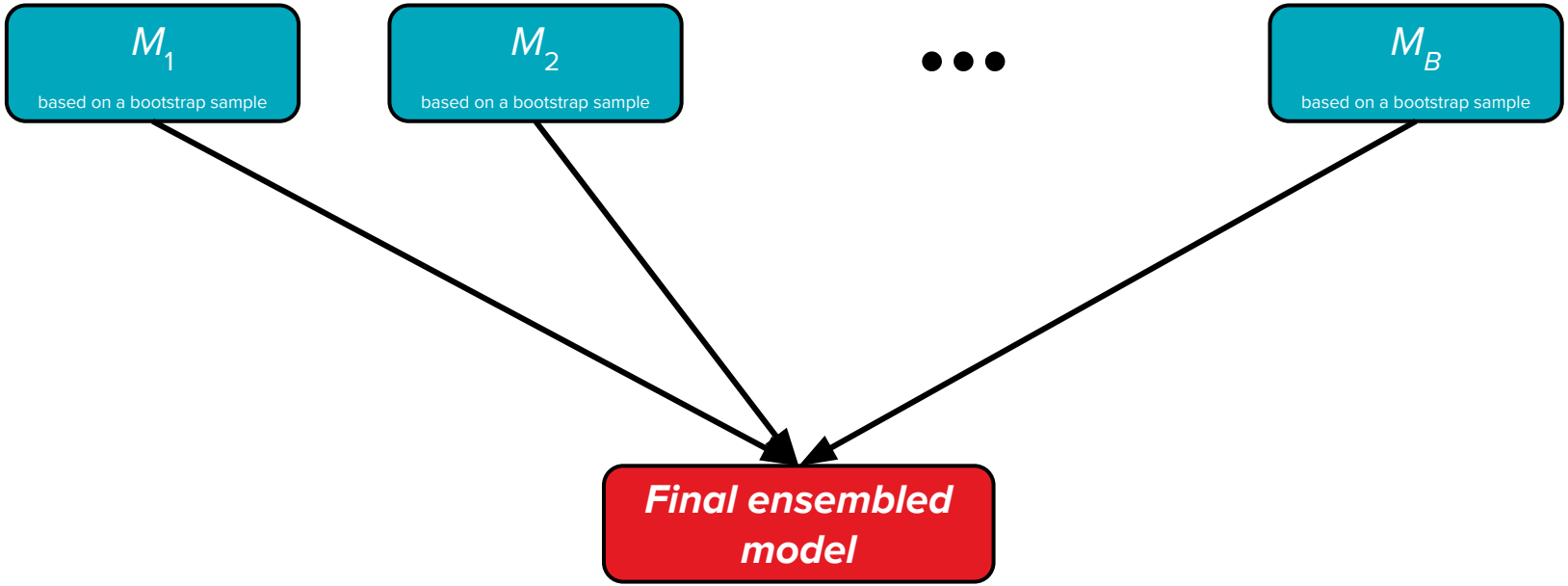
Ensembling

Ensemble Models

Bagging is one of many types of **ensembling** models together.

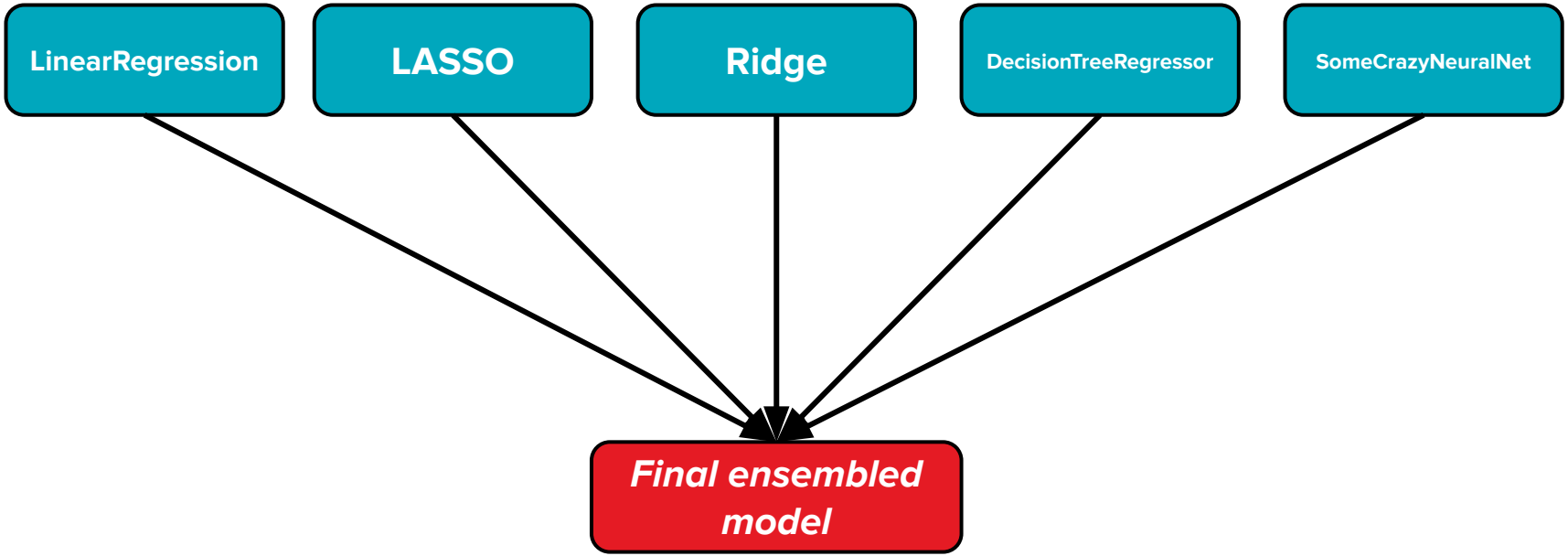
An **ensemble model** is one that is the result of combining the predictions of multiple other models.

Bagging is ensembling!



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We can also have arbitrary models ensembled together!



Let's try it out!



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